

Solution to Footnote 14

November 18, 2025

Problem 1. *Alice generated some secret primes p and q and calculated $n = pq$. But she slipped and made $f = (p - 1)(q - 1)$ public (in addition to n). Can you use n and f to recover p and q ?*

Solution. We know that $n = pq$ and $f = (p - 1)(q - 1)$. And since

$$\begin{aligned} & n - f + 1 \\ &= pq - (p - 1)(q - 1) + 1 \\ &= pq - (pq - p - q + 1) - 1 \\ &= p + q \end{aligned}$$

we know the values $A = pq$ and $B = p + q$. Knowing those values, there are multiple ways to compute p and q . One is: Since we know that the polynomial $(B - x)x$ attains its maximum at $\frac{B}{2}$ and is monotone on $\{x \mid x \geq \frac{B}{2}\}$ we can binary search on B for A . $\square$